

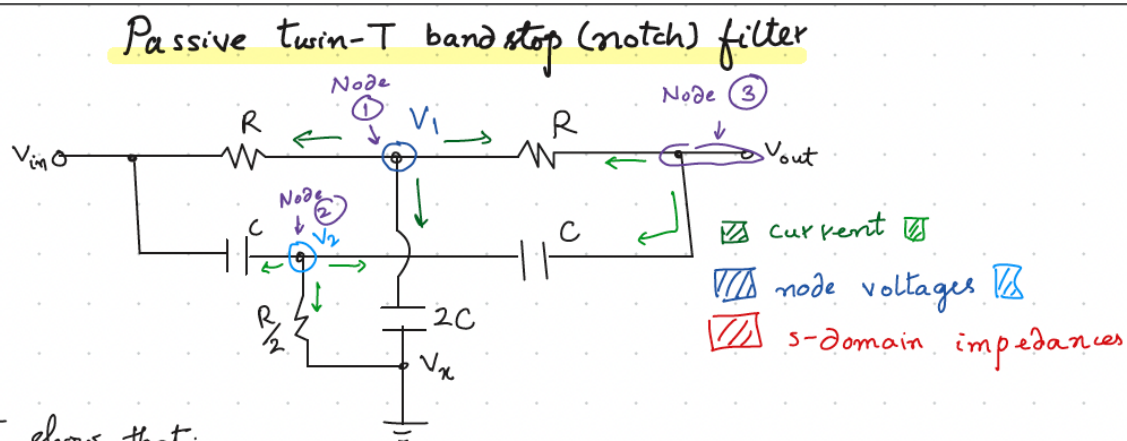
EE 202L - Linear Circuits
Final Project – The Corrupted Chord
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This report describes the design and implementation of a notch filter for a cigar-box electric guitar. The objective of the circuit was to suppress the 196 Hz fundamental of the unfretted G string while preserving nearby notes as much as possible. The design was based on a twin-T bandstop filter, later sharpened with positive feedback so that the notch became narrow enough to reject the target note without strongly attenuating adjacent frequencies.

1. Filter Design

The starting point for the design was the passive twin-T notch filter assigned in the project handout. The details for theoretical calculations and derivations are shown in the handwritten screenshots below:



We must show that:

$$\frac{V_{out}}{V_{in}} = \frac{K(s^2 + \omega_0^2)}{s^2 + \frac{5\omega_0}{Q}s + \omega_0^2} \quad \text{where } \omega_0 = \frac{1}{RC}, \quad Q = \frac{1}{4}, \quad \text{and } K = 1$$

To perform s-domain analysis we shall use impedances:

Resistors $R \rightarrow R$; $\frac{R}{2} \rightarrow \frac{R}{2}$ Capacitors $C \rightarrow \frac{1}{sC}$; $2C \rightarrow \frac{1}{2sC}$

It will help computation to express as admittances:

Resistors $R \rightarrow \frac{1}{R}$; $\frac{R}{2} \rightarrow \frac{2}{R}$ Capacitors $\frac{1}{C} \rightarrow sC$; $\frac{1}{2C} \rightarrow 2sC$

@ Node 1

KCL [currents leaving node] using node voltages yields:

$$\frac{V_1 - V_{in}}{R} + \frac{V_1 - V_{out}}{R} + 2sC V_1 = 0$$

$$\Rightarrow (V_1 - V_{in}) + (V_1 - V_{out}) + 2sRC V_1 = 0$$

$$\Rightarrow V_1 (2 + 2sRC) = V_{in} + V_{out} \Rightarrow V_1 = \frac{V_{in} + V_{out}}{2(1 + sRC)} \rightarrow \text{Eq}^n (1)$$

@ Node 2

KCL [currents leaving node] using node voltages yields:

$$sC(V_2 - V_{in}) + sC(V_2 - V_{out}) + \frac{2V_2}{R} = 0$$

$$\Rightarrow 2sC V_2 + \frac{2V_2}{R} = sC V_{in} + sC V_{out}$$

$$\Rightarrow \left(2sC + \frac{2}{R}\right) V_2 = sC(V_{in} + V_{out}) \Rightarrow V_2 = \frac{sRC(V_{in} + V_{out})}{2(1 + sRC)} \rightarrow \text{Eq}^n (2)$$

@ Node 3

$$\frac{V_{out} - V_1}{R} + sC(V_{out} - V_2) = 0$$

$$\Rightarrow V_{out} - V_1 + sRC(V_{out} - V_2) = 0$$

Substituting expressions for V_1 & V_2 using Eqⁿ ① + Eqⁿ ②

$$V_{out} - \frac{V_{in} + V_{out}}{2(1+sRC)} + sRC\left(V_{out} - \frac{sRC(V_{in} + V_{out})}{2(1+sRC)}\right) = 0$$

let $x = sRC$

Then

$$V_{out} - \frac{V_{in} + V_{out}}{2(1+x)} + x\left(V_{out} - \frac{x(V_{in} + V_{out})}{2(1+x)}\right) = 0$$

$$\Rightarrow V_{out} - \frac{V_{in} + V_{out}}{2(1+x)} + xV_{out} - \frac{x^2(V_{in} + V_{out})}{2(1+x)} = 0$$

$$\Rightarrow (1+x)V_{out} - \frac{(1+x^2)(V_{in} + V_{out})}{2(1+x)} = 0$$

$$\Rightarrow 2(1+x)^2 V_{out} - (1+x^2)(V_{in} + V_{out}) = 0$$

$$\Rightarrow [2 + 4x + 2x^2 - 1 - x^2] V_{out} = (1+x^2) V_{in}$$

$$\Rightarrow (x^2 + 4x + 1)V_{out} = (1+x^2) V_{in}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{1+x^2}{1+4x+x^2} \quad \text{substituting back } x = sRC$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{1+(sRC)^2}{1+4sRC+(sRC)^2} \quad \text{divide numerator/denom. by } (RC)^2$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{\frac{1}{(RC)^2} + s^2}{\frac{1}{(RC)^2} + \frac{4}{RC}s + s^2} \quad \text{Define } \boxed{\omega_0 = \frac{1}{RC}}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \overset{K}{\times} \frac{(s^2 + \omega_0^2)}{s^2 + \underset{\frac{1}{Q}}{4}\omega_0 s + \omega_0^2}$$

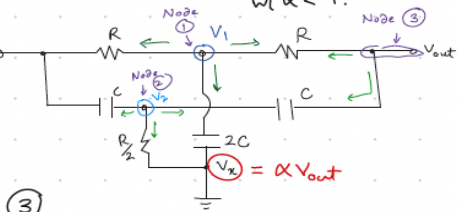
Comparing w/ given eqⁿ: $\frac{V_{out}}{V_{in}} = \frac{K(s^2 + \omega_0^2)}{s^2 + \frac{s\omega_0}{Q} + \omega_0^2}$

We get $\boxed{K=1}$, $\boxed{Q=\frac{1}{4}}$, and $\omega_0 = \frac{1}{RC}$

Because $K = 1$, the passband gain is ideally unity. The numerator becomes zero when $s = j\omega_0$, so the output ideally goes to zero at the notch frequency. Therefore, the notch frequency depends only on the product RC .

The project required the notch to occur at $f_0 = 196$ Hz. To obtain a narrow notch, the passive network was modified with positive feedback so that the center node voltage became $V_x = \alpha V_{out}$, with $\alpha < 1$. The handout states that the notch frequency remains unchanged and that the quality factor increases according to $Q = \frac{1}{4(1-\alpha)}$.

Now, we let this circuit be subject to positive feedback; i.e. $V_x = \alpha V_{out}$
 Proof that ω_0 is unchanged, but $Q = \frac{1}{4(1-\alpha)}$



KCL @ ①: $\frac{V_1 - V_{in}}{R} + \frac{V_1 - V_{out}}{R} + 2sC(V_1 - \alpha V_{out}) = 0$
 $\Rightarrow V_1 - V_{in} + V_1 - V_{out} + 2sRCV_1 - 2sRC\alpha V_{out} = 0$
 $\Rightarrow V_1(2 + 2sRC) = V_{in} + (1 + 2\alpha sRC)V_{out}$
 $\Rightarrow V_1 = \frac{V_{in} + (1 + 2\alpha sRC)V_{out}}{2(1 + sRC)} \rightarrow \text{Eq}^n \text{ ③}$

KCL @ ②: $sC(V_2 - V_{in}) + sC(V_2 - V_{out}) + \frac{2}{R}(V_2 - \alpha V_{out}) = 0$
 $\Rightarrow (2 + 2sRC)V_2 = sRCV_{in} + (sRC + 2\alpha)V_{out}$
 $\Rightarrow V_2 = \frac{sRCV_{in} + (sRC + 2\alpha)V_{out}}{2(1 + sRC)} \rightarrow \text{Eq}^n \text{ ④}$

KCL @ ③: $\frac{V_{out} - V_1}{R} + sC(V_{out} - V_2) = 0$
 $\Rightarrow V_{out} - V_1 + sRC(V_{out} - V_2) = 0$
 let $x = sRC$
 then: $V_{out} - \frac{V_{in} + (1 + 2\alpha x)V_{out}}{2(1 + x)} + x \left(V_{out} - \frac{xV_{in} + (x + 2\alpha)V_{out}}{2(1 + x)} \right) = 0$
 $\Rightarrow 2(1 + x)^2 V_{out} - V_{in} - (1 + 2\alpha x)V_{out} + xV_{out} - x^2 V_{in} - (x^2 + 2\alpha x)V_{out} = 0$
 $\Rightarrow \left[2(1 + x)^2 - (1 + 4\alpha x + x^2) \right] V_{out} = (1 + x^2) V_{in}$
 $2 + 4x + 2x^2 - 1 - 4\alpha x - x^2 = 1 + 4(1 - \alpha)x + x^2$
 $\Rightarrow \frac{V_{out}}{V_{in}} = \frac{1 + (sRC)^2}{1 + 4(1 - \alpha)sRC + (sRC)^2} \quad \left\{ x = sRC \right\}$
 $\Rightarrow \frac{V_{out}}{V_{in}} = \frac{\frac{1}{(RC)^2} + s^2}{\frac{1}{(RC)^2} + \frac{4(1 - \alpha)s}{RC} + s^2} \quad \omega_0 = \frac{1}{RC} \quad \text{unchanged}$
 $\Rightarrow \frac{V_{out}}{V_{in}} = \frac{1[s^2 + \omega_0^2]}{s^2 + 4(1 - \alpha)s\omega_0 + \omega_0^2}$
 $\Rightarrow \underline{k=1} \quad \Rightarrow \frac{1}{Q} = 4(1 - \alpha) \Rightarrow \underline{Q = \frac{1}{4(1 - \alpha)}}$

Furthermore, below is the justification for the choice of resistor and capacitor values:

Determine design values for R , C , and α s.t. $f_0 = 196 \text{ Hz}$ [G-string] & $Q = 8.5$

Using $Q = \frac{1}{4(1-\alpha)} \Rightarrow 4(1-\alpha) = \frac{1}{8.5} \Rightarrow \alpha = 1 - \frac{1}{34} \Rightarrow \alpha = \frac{33}{34} \approx 0.971$

$\omega_0 = 2\pi f_0$ & $\omega_0 = \frac{1}{RC}$

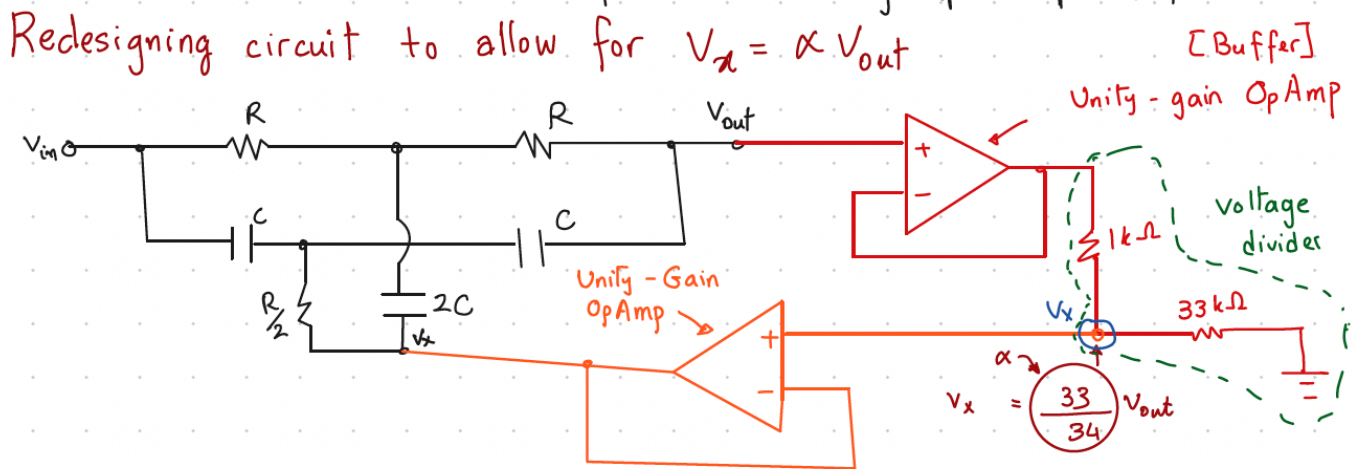
$\Rightarrow \frac{1}{2\pi RC} = f_0 \Rightarrow RC = \frac{1}{2\pi(196)} \approx 8.12 \times 10^{-4} \text{ s} \Rightarrow C = \frac{8.12 \times 10^{-4}}{10^4} = 81.2 \text{ nF}$ let $R = 10 \text{ k}\Omega$

So, for this circuit we chose $R = 10 \text{ k}\Omega \Rightarrow R/2 = 5 \text{ k}\Omega$ as 

and $C = 81.2 \text{ nF}$ as 

and $2C = 162.4 \text{ nF}$ as two of these 81.2 nF groups in parallel

The positive-feedback factor alpha was produced with a voltage divider and two unity-gain buffers (OpAmps), as shown below:



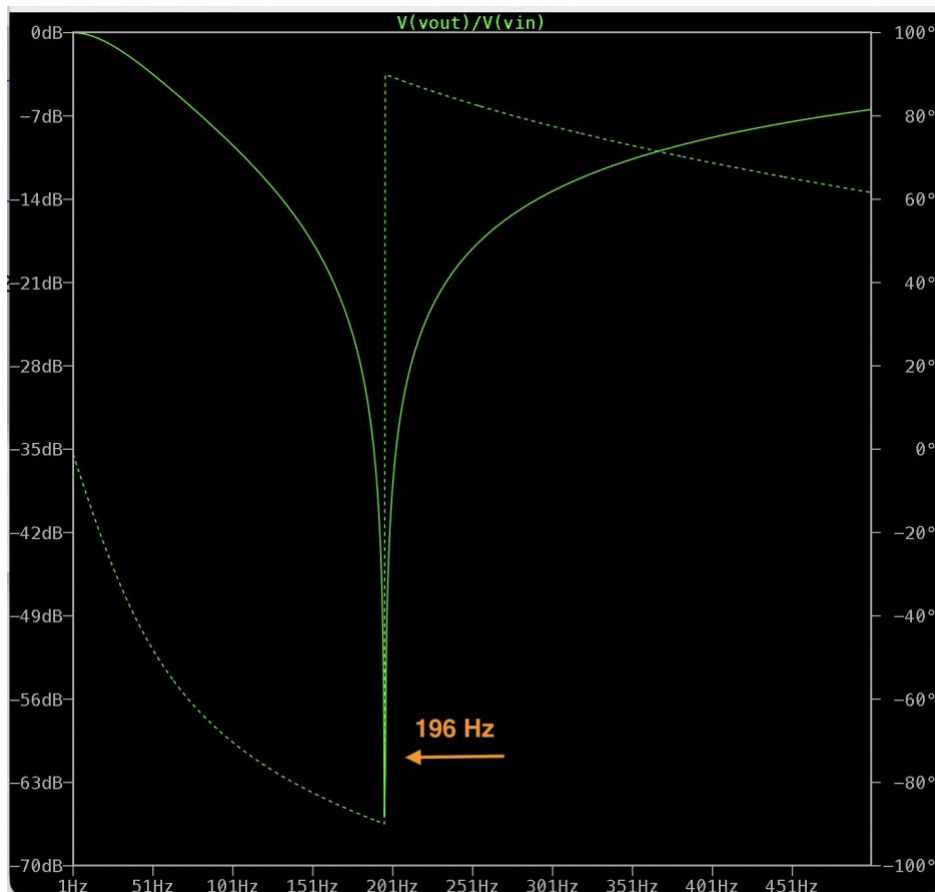
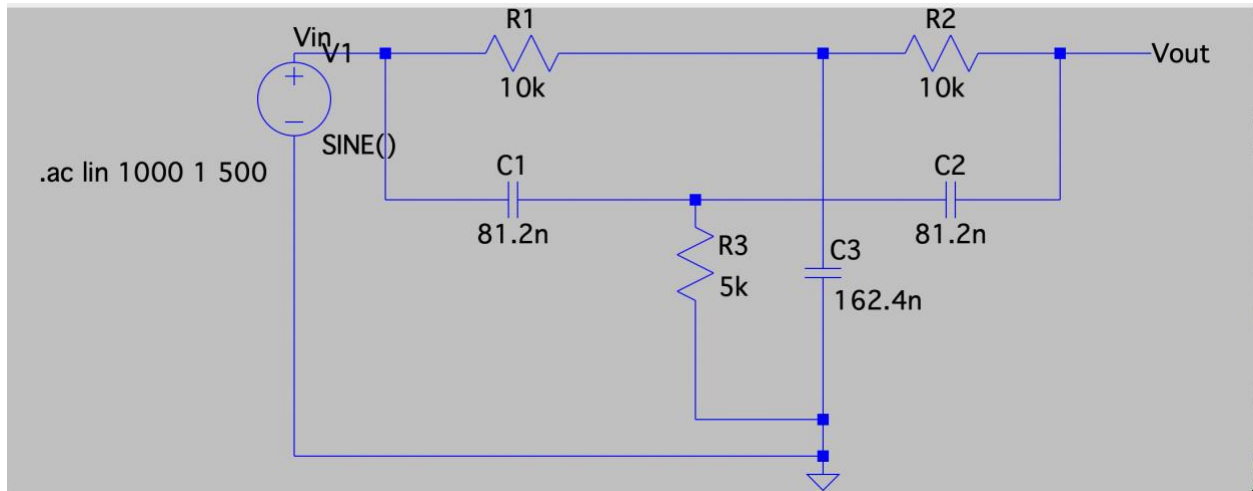
With these values, the theoretical notch frequency is 196 Hz and the theoretical quality factor is 8.5. This satisfies the project requirement that the filter reject the G-string fundamental while keeping the bandwidth narrow enough that nearby notes are minimally affected.

2. Simulations

We performed an AC simulation of the notch filter using a linear frequency sweep. A linear sweep is appropriate here because the magnitude changes rapidly in a relatively narrow

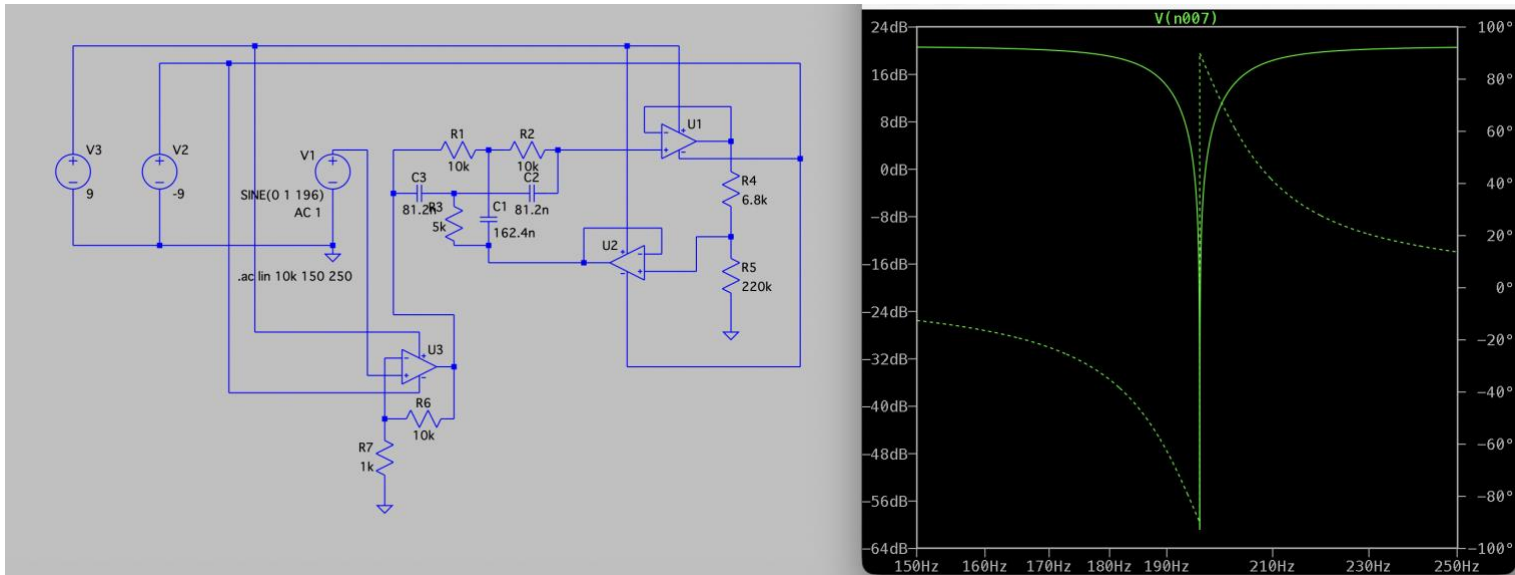
frequency range around the notch, so a logarithmic sweep can make the dip look less clear. The specified AC source format was `Vin 1 0 ac 1m`, which sets the small-signal AC input amplitude to 1 mV.

The sweep command format was `.ac LIN N Start Stop`, followed by `.probe`. In the probe window, the plotted transfer function should be `Vout/Vin`.



We clearly see the 'bandstop' or notching action of the filter here. Whilst LTSpice on Mac did not allow exact point labelling, we could trace around the drop to confirm that this notch filter effectively notched out the expected frequency of 196 Hz.

For the active version of the circuit, we built the circuit below:



The expected result is a nearly flat passband close to 0 dB away from the notch, together with a deep, narrow dip centered near 196 Hz. Since the ideal transfer function has unity gain in the passband, the simulated response outside the reject band should remain near 1 in linear scale. At the notch frequency, the ideal output should approach zero, which appears as a very large attenuation in the magnitude plot.

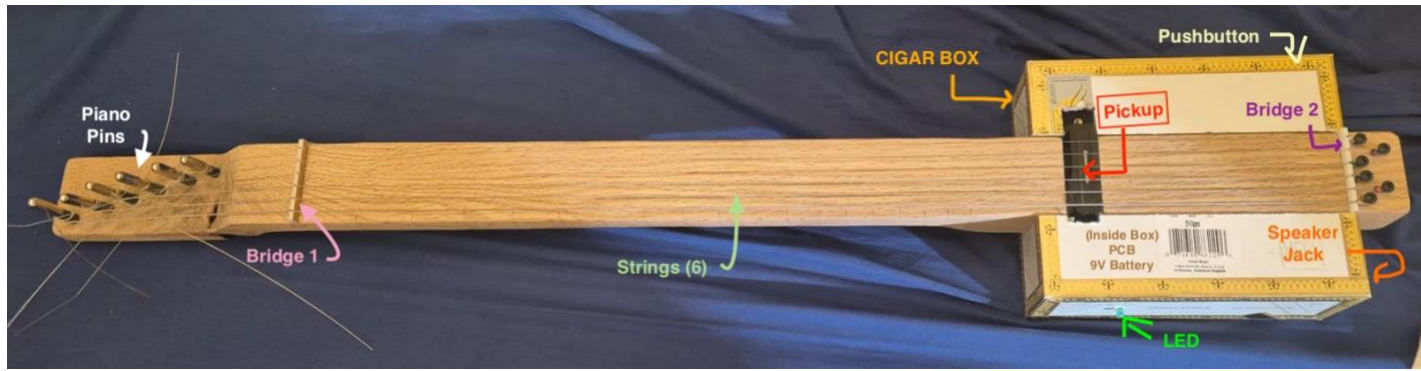
The target quality factor is $Q = 8.5$, so the notch should be relatively narrow. Using $BW = f_0 / Q$ gives an approximate bandwidth of $196 / 8.5 = 23.1$ Hz. This means the simulated attenuation should be concentrated in a small region around 196 Hz rather than extending broadly across the surrounding notes. When a component is shifted slightly from its nominal value, the expected response is that the notch moves away from 196 Hz and becomes less deep, which explains why component matching and post-build tuning are important.

The handout also states that the measured attenuation at f_0 relative to the passband should be at least -20 dB. In ideal simulation, the notch will usually be deeper than this requirement, but practical component mismatch will reduce the achievable depth in the real circuit.

3. Guitar Assembly

The physical build of the project centered on converting the supplied wood and cigar box into a functional electric guitar body that could support the electronic filter board. We took a ‘standard guitar’ wood piece. During assembly, the pickup, bridge, strings, screws, and tuning hardware were installed so that the string path passed properly over the pickup.

Below is a labelled photograph of our guitar:

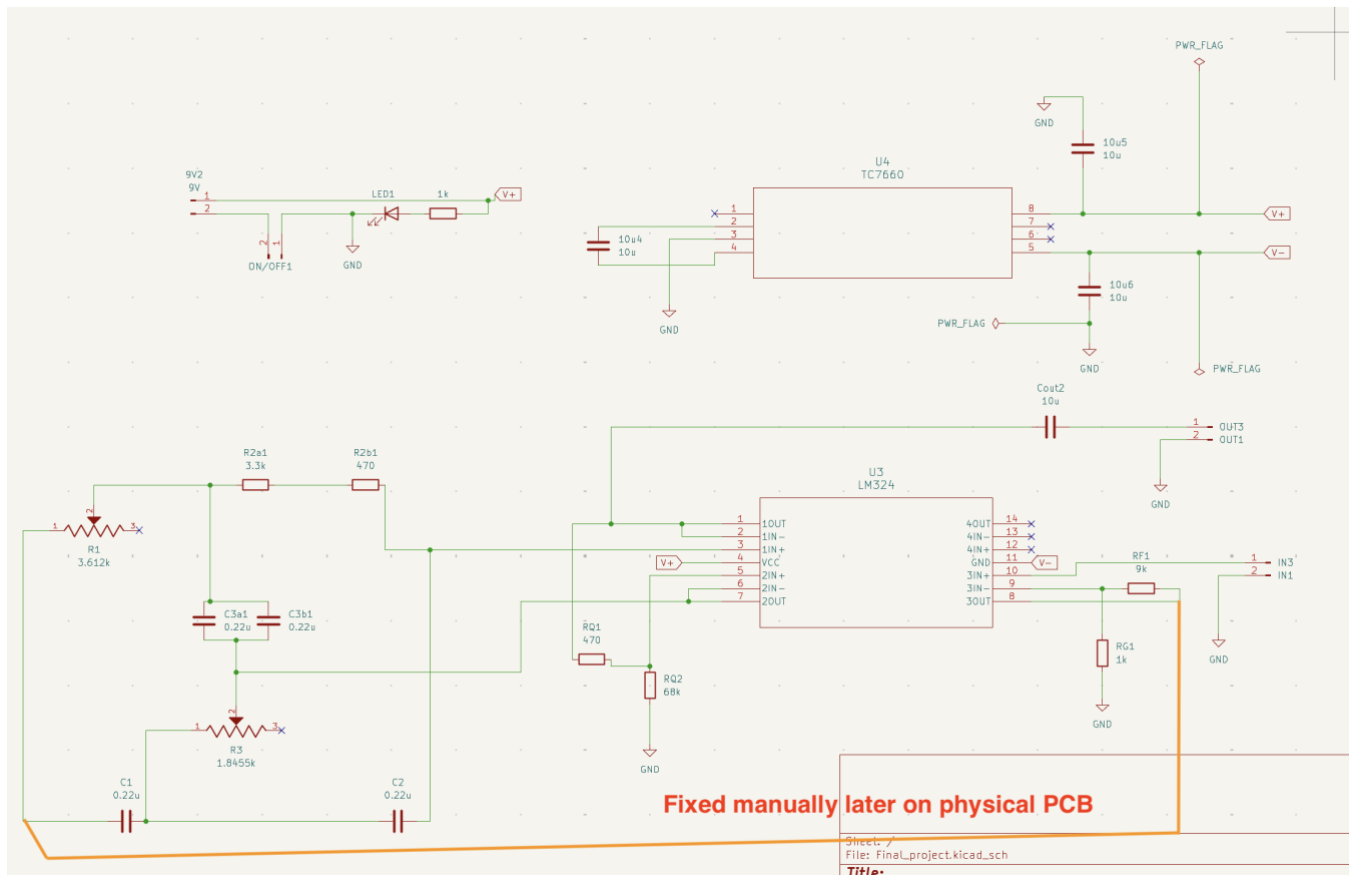


The final assembly stage required combining the mechanical and electrical portions of the project into a single working system. This included checking string tension, verifying that the pickup was mounted securely, and ensuring that the PCB and 9V battery were secured well inside the box. A successful mechanical build made the later electrical testing much easier because it reduced the number of non-electrical variables that could interfere with the checkoff.

4. PCB Design

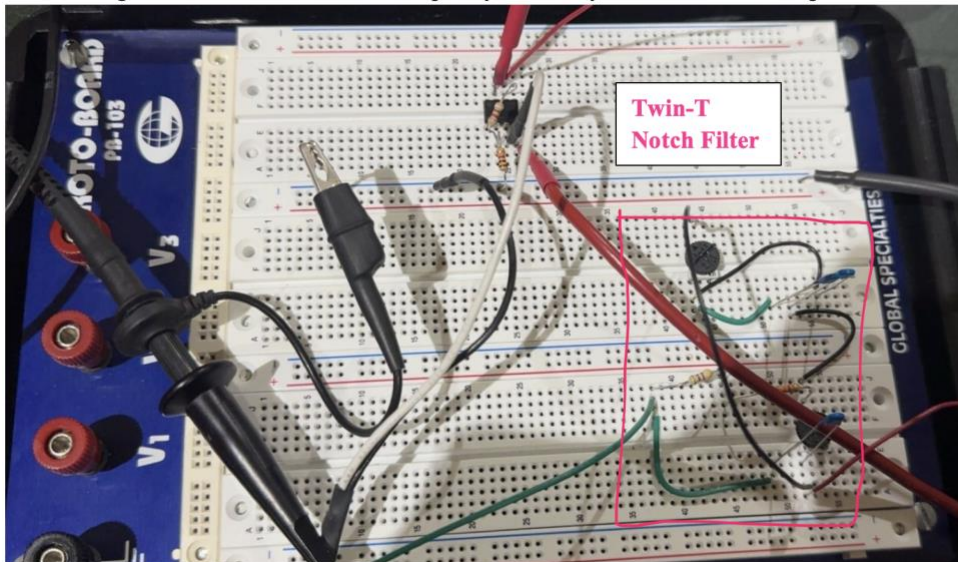
After the filter concept was confirmed on the breadboard, the circuit was transferred into KiCad for schematic capture and PCB layout. The schematic included the twin-T network, the positive-feedback divider, the op-amp buffers required to isolate the divider from the filter nodes, and the necessary power and connector interfaces for the pickup, battery, and speaker path.

Shown below is our schematic:

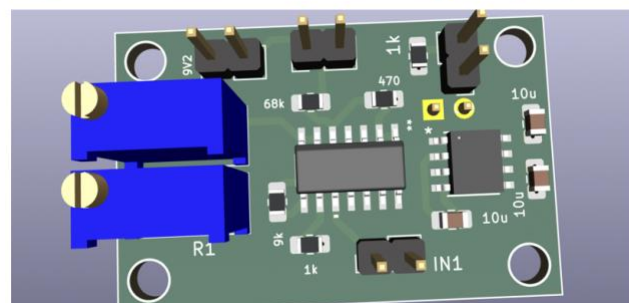
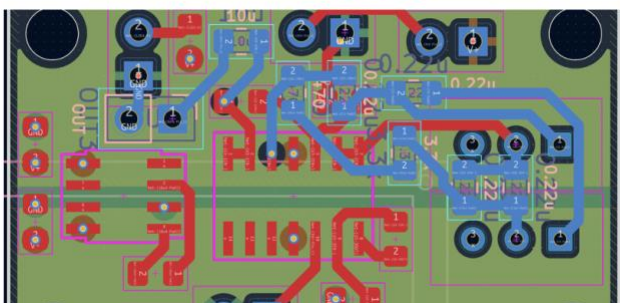
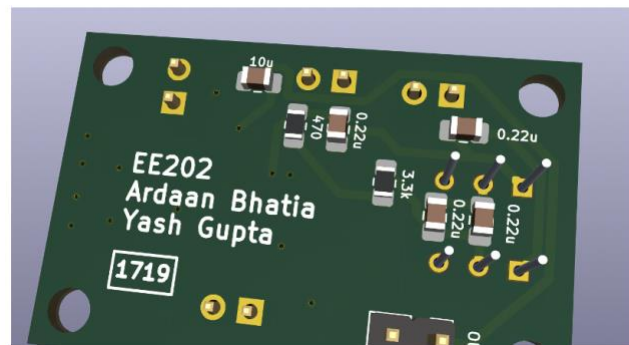
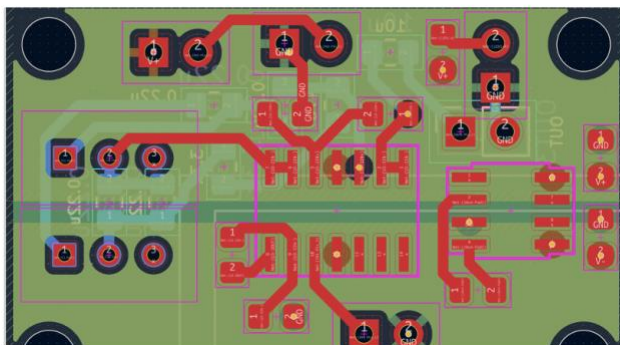


As seen by the **orange wire**, we didn't have it originally, and later discovered this mistake post PCB fabrication. Fortunately, we could just manually solder on an additional wire to the pads.

Moving from breadboard to PCB required verifying every connection carefully because even a small wiring error could shift the notch frequency or destroy the desired filter shape.



Component placement on the PCB was chosen to keep the filter network organized and to make soldering and debugging practical. The passive components associated with the notch network were grouped closely to reduce wiring confusion, while the active devices and power connections were arranged so that probing and rework remained manageable during testing. Clear labeling of polarity-sensitive parts was important for efficient, error-free soldering.



Once fabricated, the PCB was populated by soldering the op-amps, resistors, capacitors, and connectors onto the board. The assembled board was then checked for shorts, missing joints, incorrect orientations, and other common manufacturing or soldering defects before integration into the guitar. This stage served as the bridge between the theoretical design and the final functional instrument.

5. Final Electrical Testing

The completed PCB was installed inside the cigar-box body and connected to the pickup, speaker, and battery supply so that the full guitar system could be tested as an integrated design. In particular, the pickup output (hot black line) was soldered onto 'IN' pins of PCB and the 'OUT' pins of the PCB were wired to the speaker jack. Electrical testing focused on confirming that the active notch filter still behaved correctly after installation and that the instrument produced a clear output through the speaker.

The main performance checks were the notch frequency f_0 , the quality factor Q , and the attenuation at the target note relative to the passband. The expected result was a clear suppression of the 196 Hz G-note fundamental together with acceptable transmission of nearby notes, demonstrating that the notch was narrow enough for the intended application. To achieve this, we tuned the potentiometers. This behavior's demonstration video is included in the Brightspace submission as 'Yash_Ardaan_NotchFilter_Demo.mp4'.

Then, we finally got to try out the playing the guitar, it worked fluently when connected to the speaker. Whilst we observed that the G is audible, in spite of oscilloscope confirmation, it made sense due to harmonics and resonance associated with playing a guitar string. A successful final test therefore confirmed not only the circuit equations and simulations, but also the practical integration of the electronics into the guitar platform. A video demonstration named 'Yash_Ardaan_Guitar_Demo.mp4' is also attached in the Brightspace submission.